Fiscal Year:	FY 2024	**Task Last Updated:**	FY 01/19/2024
PI Name:	Yagoobi, Jamal Ph.D.		
Project Title:	Electrically Driven Liquid Film Flow Boiling: A Two-Phase Heat Transport Device Driven By Electric Conduction Mechanism		
Division Name:	Physical Sciences		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	jyagoobi@wpi.edu	**Fax:**	FY
PI Organization Type:	UNIVERSITY	**Phone:**	508-831-5759
Organization Name:	Worcester Polytechnic Institute		
PI Address 1:	Department of Mechanical Engineering		
PI Address 2:	10 W 32nd St, E1 Building, Room 243		
PI Web Page:			
City:	Worcester	**State:**	MA
Zip Code:	01609-2280	**Congressional District:**	3
Comments:	NOTE PI formerly at Illinois Institute of Technology; moved to Worcester Polytechnic Institute in January 2012. Info per PI (Ed., 1/17/2012)		
Project Type:	Flight,Ground	**Solicitation / Funding Source:**	Physical Sciences Unsolicited
Start Date:	03/01/2022	**End Date:**	02/28/2026
No. of Post Docs:		**No. of PhD Degrees:**	1
No. of PhD Candidates:	2	**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA GRC
Contact Monitor:	Hasan, Mohammad	**Contact Phone:**	216-977-7494
Contact Email:	Mohammad.M.Hasan@nasa.gov		
Flight Program:	ISS		
Flight Assignment:	ISS		
Key Personnel Changes/Previous PI:	Not applicable		
COI Name (Institution):			
Grant/Contract No.:	80NSSC22K0676		
Performance Goal No.:			
Performance Goal Text:			
	NOTE: Task Description updated from original proposal (Ed., 7/18/23) The basic science and experimental effort covered by NASA Grant/Cooperative Agreement Number: NNX16AT09G entitled, “Electrically Driven Liquid Film Boiling - A Two-Phase Heat Transport Device Driven by Electric Conduction Mechanism” is essentially continuing with this new proposal. The specific tasks of the proposed project are: Task 1: Support the final design and analysis of the experiment conduction disc and heater, as well as detailed thermal analysis of the thermodynamics governing the experiment chamber. Develop and conduct ground based experiment to determine the final International Space Station (ISS) experimental flight matrix.		

Task Description:	Task 2: Complete tests as stated in the Science Requirements Document (SRD) (period: October 2021 to December 2022), with 2 mm liquid film thickness. The same will be conducted with the liquid film thickness of 3 mm. Upon completion of this effort, the combined results of 2 mm and 3 mm tests will used to finalize the test matrix to be conducted aboard the ISS. Task 3: Investigate the impact of the presence of non-condensable gas. The dielectrophoretic (DEP) force extracts the vapor bubble away from the heat surface. This is because of the difference between the vapor phase and liquid phase electric permittivity values. The vapor with a lower electric permittivity is attracted toward a lower electric field (i.e., DEP electrode) and the liquid with a higher electric permittivity is pushed toward the higher electric field (i.e., the heater). In the presence of a non-condensable gas (i.e., air), the DEP will extract the air away from the heater surface, thus minimizing or removing the impact of non-condensable gas on the liquid/vapor phase change taking place at the heater surface. A series of tests will be conducted at Worcester Polytechnic Institute (WPI) by introducing air at different levels. The corresponding Psat and Tsat will be measured to determine the amount of air introduced into the test vessel. In conclusion, the DEP force not only enhances the boiling and makes it feasible in the absence of gravity; it is also expected to eliminate the negative impact of the presence of non-condensable gas in the working fluid. Task 4: Numerical modelling of electrically driven liquid film flow in the presence of phase change. The electrically driven liquid film flow generated by the electrohydrodynamic (EHD) conduction mechanism will be mathematically modeled to provide scientific results for the corresponding flow field and electric field. The problem in hand is very unique and challenging since the liquid film thickness changes in the direction of the flow due to condensation taking place at its liquid/vapor interface. Furthermore, the downstream boundary condition will need to take into account the presence of boiling and the DEP vapor extraction force. The numerical results will be compared in the presence and absence of gravity. Task 5: Detailed flow field measurements of electrically driven liquid film flow. The electrically driven liquid film flow field will be measured with the particle imaging velocimetry (PIV) unit. This will provide valuable data which will be used to compare them to the numerical predictions stated in Task 3. Initial PIV experiments will be conducted using our linear two-phase flow setup. Afterwards, modifications will be made to the radial unit, followed by corresponding flow field measurements with the PIV unit. Task 6: A detailed long-term study will be conducted to evaluate the system performance with respect to time and make sure that the effects of EHD conduction pumping of liquid film and the extraction of vapor bubbles do not degrade over time. Tentatively, a period of a full year will be considered for this continuous operation study.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The rapid development of modern electronics has resulted in significant computational power in increasingly small packaging. However, a significant challenge remains in effectively controlling the temperature of electronic components, especially in harsh environments including space. With the revitalized interest in space development through NASA's Artemis Program and the increasing commercialization of space, development of improved thermal management technologies is needed with the objectives of reducing the mass and increasing the efficiency of heat transfer systems. Electrohydrodynamic (EHD) pumps have been identified as an advanced technology to support NASA's future human, robotic, and deep space missions. Electrohydrodynamic conduction pumps in particular have the advantage of being lightweight, with no moving parts, and are highly effective for thermal enhancement. From the progress made in this project, the body of knowledge and understanding of EHD conduction pumping applied for thermal management of electronics on earth as well as in space has been expanded. Though this research is targeted for electronics cooling in space, utilization of EHD conduction pumping for enhancement of existing terrestrial systems, such as vapor-compression cycle refrigeration, has the potential for improving the overall system efficiency. There is also the potential to improve the thermal performance of electronics, such as those in data centers found throughout the world. Thus, the future implementation of EHD has the potential for significant cost savings, and reduction in carbon footprint.
Task Progress:	The boundaries of knowledge in the field of electrohydrodynamic (EHD) conduction driven thermal management for space has been extended through this project. EHD has been demonstrated as an effective method for flow distribution control in a parallel microchannel evaporator, capable of redistributing flow at heat fluxes up to 77.5 W/cm2. Unique EHD conduction pumping design was developed for generating swirl flow to improve heat transfer. This design was numerically modeled and validated against experimental static pressure generation data. Pumps were fabricated through additive manufacturing methods, and their heat transfer enhancement capabilities were demonstrated, showing up to a 70% increase in local heat transfer coefficient. The effect of working fluid temperature on the performance of two different pumps has been presented to validate the theoretical model of EHD conduction, and demonstrated the ohmic and saturation regimes of EHD conduction pumping, which is critical for implementation on future spacecraft and satellites experiencing changing temperature conditions. The transient behavior was also seen to be a function of temperature. The effect of non-condensable gas concentration on electrically driven liquid film flow boiling was explored and it was demonstrated that the application of electric fields can negate the detrimental heat transfer effects. Finally, the effect of pulsating voltage application to the electrically driven liquid film flow boiling system was explored, and increases in single phase heat transfer and boiling critical heat flux were observed. Surface waves generated at the liquid-vapor interface were found to be in either shallow or deep regimes, depending on the frequency. The frequency of pulsation significantly influenced the flow behavior of the liquid film. While the presented research has been specifically targeted for space applications, the impacts of this work on society are far reaching. Development of next-generation EHD conduction-based thermal management technologies will impact terrestrial life in many ways.
Bibliography Type:	Description: (Last Updated: 01/30/2026)

Articles in Other Journals or Periodicals	O'Connor N, Talmor M, Yagoobi J. "Electrohydrodynamic flow distribution control for parallel microchannel evaporators. " Available at SSRN preprint server. Abstract 4379487. Posted March 6, 2023. http://dx.doi.org/10.2139/ssrn.4379487 , Mar-2023
Articles in Peer-reviewed Journals	Castaneda AJ, O'Connor NJ, Yagoobi JS, Didion JR, Martins MS, Hasan MM. "Dielectrophoretically-assisted electrohydrodynamic-driven liquid film flow boiling in the presence and absence of gravity." ASME Journal of Heat and Mass Transfer. 2023 Mar 1;145(3):031601. https://doi.org/10.1115/1.4055566 , Mar-2023
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Dissertations and Theses	O'Connor N. "Electrohydrodynamically driven thermal management for space and terrestrial applications – An experimental and numerical study." Dissertation, Worcester Polytechnic Institute, August 2023. , Aug-2023
Papers from Meeting Proceedings	Castaneda AJ, Yagoobi J. "Experimental and numerical study of an oscillating rlectrohydrodynamic conduction pump." 22nd Conference on IEEE International Conference on Dielectric Liquids (ICDL), Worcester, MA, June 25-28, 2023. E-Proceedings. 22nd Conference on IEEE International Conference on Dielectric Liquids (ICDL), Worcester, MA, June 25-28, 2023. , Jun-2023
Papers from Meeting Proceedings	O'Connor N, Yagoobi J. "The effect of pulsating dielectrophoresis on liquid film flow boiling." 22nd Conference on IEEE International Conference on Dielectric Liquids (ICDL), Worcester, MA, June 25-28, 2023. E-Proceedings. 22nd Conference on IEEE International Conference on Dielectric Liquids (ICDL), Worcester, MA, June 25-28, 2023. , Jun-2023